MTH202 MID TERM PAST PAPER BY SAQIB

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MCQS:

x^2-9|=…….

►|(x-3)^2|
►|(x+3)^2|
►***\|x-3\|\|x+3\|***
►|x+3||x+3|

Usually the number that signifies the idea of f(x) being as close to limit L as want to be must be a/an ……

►***Integer***
►Natural number
►Small positive number.
►Small negative number.

A function f is said to be continuous on a closed interval [a, b] if f is continuous from the right at “a” and “f” is continuous from the left at “b” and “f” is continuous on

►(a,b]
►[a,b)
►[a,b]
►***(a, b)***

If f is continuous on [a, b], and if f(angel) and f(beer) have opposite signs, then there is …… one solution of the equation f(x) = 0 in the interval (a, b).

►at most
►exactly
►***at least***
►not more than

e (epsilon) used in the definition of limit can be a negative number.

►True
►***False***

If a function is differentiable at a point then it is continuous at that point. The converse is

►False
►***True***

If the function f and g are continuous at c, then f + g is …… at c.

►Discontinuous
►***Continuous***

If f is continuous on a closed interval [a, b] and C is any number between f(angel) and f(beer), inclusive, then there is at least one number x in the interval [a, b] such that ---------

►f(x) is not equal to C
►***f(x) = C***
►f(x)>C
►f(x)<C

|x-3| < 1 implies.....

►-4 < x < 4
►***2 < x < 4***
►-2 < x < -4
► x-3 < 1

If for any positive number e(epsilon) we can find d (delta) such that| (3x-5) - 1| < e ,if x satisfies 0< |x-2| < d Then f(x) =……

►3x-5- 1

►x-2

►***3x-5***

►None of these

Question # 1 of 10 (Start time: 05:59:11 PM) Total Marks: 1
A graph of an equation is the points on the xy- plane that ……the equation.
Select correct option:

►***Satisfy***
►Does not satisfy.

Question # 2 of 10 (Start time: 06:00:12 PM) Total Marks: 1
If two lines have same slope, say 1, then these two lines are _____
Select correct option:

►***Parallel to each other***
►Parallel to x-axis
►Parallel to y-axis
►Perpendicular to each other

Question # 3 of 10 (Start time: 06:00:57 PM) Total Marks: 1
There is one-to-one correspondence between the points on co-ordinate line and _____
Select correct option:

►Set of natural numbers
►Set of integers
►Set of irrational numbers
►***Set of real numbers***

Question # 4 of 10 (Start time: 06:02:06 PM) Total Marks: 1
Which of the following is y-intercept of a line represented by equation, 5y = - 2x + 16
Select correct option:

►2
►5
►8
►16 / 5

Question # 5 of 10 (Start time: 06:03:11 PM) Total Marks: 1
Does the point (1,2) satisfies the equation: 2y=x2−1
Select correct option:

►Yes
►No

Question # 6 of 10 (Start time: 06:03:47 PM) Total Marks: 1
Let x is a real variable. The inequality -5 < x < 2 denote _____
Select correct option:

►Set of all rational numbers between -5 and 2
►Set of all real numbers between -5 and 2.
►Set of all natural numbers between -5 and 2
►Set of all integers between -5 and 2

Question # 7 of 10 (Start time: 06:05:01 PM) Total Marks: 1
Which of the following option describes the graph of equation:(x2)+(y2)=1
Select correct option:

►A parabola whose vertex is at point (1, 0).
►A straight line whose slope and y-intercept is 1.
►A circle centered at origin and its radius is 1
►None of these

Question # 8 of 10 (Start time: 06:05:55 PM) Total Marks: 1
Graph of the equation x2+y2=9 represents a.....
Select correct option:

►Circle not sure
►Parabola
►Ellipse
►None of these.

Question # 9 of 10 (Start time: 06:06:16 PM) Total Marks: 1
Which of the following is distance between the points (2, 5) and (-1, 1)?

Select correct option:

►Square root of (-7)
►Square root of (37)
►25
►5

Question # 10 of 10 (Start time: 06:07:05 PM) Total Marks: 1
The inequality, {1 / x} < 1 can be simplified to which of the following?
Select correct option:

► (1 - x) / x > 0
► (1 - x) / x < 0
► (1 – x) < 0
► (1 – x) > 0

If Sin(3x^2) / 6 + C is the anti–derivative of a function f(x), then f(x) = _____.

► x^2 Cos(3x^2).
► x Cos(3x^2).
► x Cos(3x).
► none

Which of the following is the integral of Sin(3x+5) with respect to x ?

► -1/3[-Cos(3x+5)]
► 1/3[-Cos(3x+5)]
► 1/15[-Cos(3x+5)]
► -Cos(3x+5)

If ‘n’ goes from 1 to 3 and the summation of ‘na’ = 6a, then the value of ‘a’ is -------

► 6
► -6
► 1
► undetermined

If ‘n’ goes from 1 to any large ODD number then the summation of ‘(-1)^n’ = -------

► ***-1***
► 0
► 1
► that specific large ODD number

1+2+3……+t equals

► n(n+1)/2
► ***t(t+1)/2***
► n(n+1)(2n+1)/6
► none

If definite integral of f(x)=Sinx over [a,0] is equal to '-2' then the value 'a' is--------

► pi/2
► ***pi***
► 0
► -pi

If the definite integral of f(x)=3 over [1,x] is greater than '12' then -----

► x>12
► x>5
► ***x>3***
► x>1

If [-8,8] is subdivided into '16' equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be--------.

► 2
► 3
► 4
► ***5***

Which of the following is the integral of sin(2x)?

► cos(2x)+C
► 2cos(2x)+C
► ***–(1/2)cos(2x)+C***
► none

Sum of cubes of n-terms of a series whose nth term is ‘n’ = ---

► Square of n(n+1)(2n+1)/6
► Square of n(n+1)/2
► ***Square of (n+1)/2 (just a guess)***
► Square of n(n+1)/6

In a rectangular coordinate system the coordinate exes divide the plane into four regions called…

► Cube
► ***Quadrants***
► Circle

An equation of the form is called ….

► Parabolic
► Cubic
► ***Quadratic***

Graph of f(x) =|x| is ……

► Circle shape
► ***V shaped***
► U shaped

Y=f(x) +3 translation of the function is at 3 is ….. By c units

► ***Up***
► Down
► Left
► Right

Graph of y=|x-5| …… of the graph y=|x| 5 units to the right to get the graph of y.

► Reading
► ***Translation***
► Creating
► Definition

By the definition and …… be on the interval (xo, x1)

► ***Greater than zero***
► Equal to zero
► Less than zero
► Less than and equal to zero

Cot(x) discontinuous at …….

► Cos(x) =0
► ***Sin(x) = 0***
► Tan(x) =0
► Cot(x) = 0

If a function is differentiable then the function is also …… on that point

► Discontinues
► No exit
► ***Continuous***

Tan (5-3x) find differentiation w.r.t. x

► ***-3sce² (5-3x)***
► 3sce² (5-3x)
► 3cos (5-3x)
► 3cos² (5-3x)

Derivative of the will be solved by …….. Method

► ***Implicit***
► Explicit
► Simple

Constant function has the slope …..

► Positive
► Negative
► ***Zero***

Decreasing function has the slope …….

► ***Negative***
► Positive
► Zero

The equation of line of the form is known as

► Point-slope form
► Two points form
► Intercepts form
► ***Slope intercept form***

The set of rational number is a subset of

► ***Integers***
► Natural numbers
► Odd integers
► Real number

If Sin(3x^2) / 6 + C is the anti–derivative of a function f(x), then f(x) = _____.

►x^2 Cos(3x^2).
►***x Cos(3x^2).***
►x Cos(3x).
►none

Which of the following is the integral of Sin(3x+5) with respect to x ?

►-1/3[-Cos(3x+5)]
►***1/3[-Cos(3x+5)]***
►1/15[-Cos(3x+5)]
►-Cos(3x+5)

If ‘n’ goes from 1 to 3 and the summation of ‘na’ = 6a, then the value of ‘a’ is ----------

►***6***
►-6
►1
►undetermined

If ‘n’ goes from 1 to any large ODD number then the summation of ‘(-1)^n’ = ---------

►***-1***
►0
►1
►that specific large ODD number

1+2+3……+t equals

►***n(n+1)/2***
►t(t+1)/2
►n(n+1)(2n+1)/6
►none

6.If definite integral of f(x)=Sinx over [a,0] is equal to ‘-2’ then the value ‘a’ is---------

►pi/2
►***pi***
►0
►-pi

If the definite integral of f(x)=3 over [1,x] is greater than ‘12’ then -----

►x>12
►x>5
►***x>3 (not sure)***
►x>1

If [-8,8] is subdivided into ‘16’ equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be--------.

►2
►3
►4
►***5***

Which of the following is the integral of sin(2x)?

►cos(2x)+C
►2cos(2x)+C

►***–(1/2)cos(2x)+C***
►none

Sum of cubes of n-terms of a series whose nth term is ‘n’ = ---

►Square of n(n+1)(2n+1)/6
►Square of n(n+1)/2
►***Square of (n+1)/2 (just a guess)***
►Square of n(n+1)/6

In a rectangular coordinate system the coordinate exes divide the plane into four regions called…

►Cube
►***Quadrants***
►Circle

An equation of the form $y = ax^2 + bx + c$ is called ….

►Parabolic
►Cubic
►***Quadratic***

Graph of f(x) =|x| is ……

►Circle shape
►***V shaped***
►U shaped

Y=f(x) +3 translation of the function is at 3 is ….. By c units

►***Up***
►Down
►Left
►Right

Graph of y=|x-5| …… of the graph y=|x| 5 units to the right to get the graph of y.

►Reading
►***Translation***
►Creating
►Definition

By the definition ε and δ …… be on the interval (xo, x1)

►***Greater than zero***
►Equal to zero
►Less than zero
►Less than and equal to zero

Let f(x) is the function as x approaches to 0 then the limit from the positive will be …

$\lim_{x \to +0} x$
$\lim_{x \to -0}$
$\lim_{x \to 0}$

Cot(x) discontinuous at …….

►Cos(x) =0
►***Sin(x) = 0***
►Tan(x) =0
►Cot(x) = 0

If a function is differentiable then the function is also …… on that point

►Discontinues
►No exit
►***Continuous***

Tan (5-3x) find differentiation w.r.t. x

►***-3sce² (5-3x)***
►3sce² (5-3x)
►3cos (5-3x)
►3cos² (5-3x)

Derivative of the $7y^4 + x^3y + x = 4$ will be solved by …….. Method

►***Implicit***
►Explicit
►Simple

Constant function has the slope …..

►Positive
►Negative
►***Zero***

Decreasing function has the slope …….

▶ ***Negative***
▶ Positive
▶ Zero

Consider two function $f(x) = x^2 and g(x) = \sqrt{x}$ **then** $fog(x) =$ **.....**

▶ x
▶ x^2
▶ $\sqrt{x}$
▶ None of these

A function is said to be continuous function if the function is continuous on the interval

▶ $(-\infty, +\infty)$
▶ $(0, +\infty)$
▶ $(-\infty, 0)$
▶ None of these

According to Power-Rule of differentiation, if $f(x) = x^n$ **where** n **is a real number, then**

$\frac{d}{dx}[x^n] =$

▶ x^{n-1}
▶ $n\,x^{n-1}$
▶ $n\,x^{n+1}$
▶ $(n-1)x^{n+1}$

30^0 = _____

▶ $\frac{\pi}{3}$
▶ $\frac{\pi}{4}$

► $\frac{\pi}{6}$

$\frac{\pi}{2}$
►

π is a .....number

► Integer
► Rational
► ***Irrational***
► Natural

The equation of line of the form $y - y_1 = m(x - x_1)$ is known as

► ***Point-slope form***
► Two points form
► Intercepts form
► Slope intercept form

The set of rational number is a subset of

► ***Integers***
► Natural numbers
► Odd integers
► Real number

Question # 1 of 10 (Start time: 11:35:52 AM) Total Marks: 1
If y=f(x) then instantaneous rate of change of y with respect to x at the point x0 is given by slope of tangent line at (x0,f(x0))
Select correct option:

►True
►False

Question # 2 of 10 (Start time: 11:37:09 AM) Total Marks: 1
If a secant line is drawn between two points P and Q on a curve, then the slope of this secant line is
Select correct option:

►=f(x2)-f(x1)
►=f(x2)-f(x1)/x2

►={f(x2)-f(x1)}/(x2-x1)
►=f(x1)-f(x2)/x2-x1

Question # 3 of 10 (Start time: 11:38:36 AM) Total Marks: 1
For the curve f(t) = -23t+25, the slope of the tangent line at point 't = -1' is
Select correct option:

►23
►-23
►25
►-25

Question # 4 of 10 (Start time: 11:40:02 AM) Total Marks: 1
If (0,0) and (2,2) are any two pints on a curve then the slope of a line parallel to the secant segment through these points is
Select correct option:

►1
►-1
►2
►-2

Question # 5 of 10 (Start time: 11:41:26 AM) Total Marks: 1
On the straight line, the tangent at any point coincide with line -------
Select correct option:

►no where
►every where
►some where
►none of these

Question # 6 of 10 (Start time: 11:42:54 AM) Total Marks: 1
The process of finding the derivative is called_____
Select correct option:

►Differentiation
►Integration
►Function
►None of these

Question # 7 of 10 (Start time: 11:43:56 AM) Total Marks: 1
If f(x) =x^3 then f '(2)= NOTE: x^n means 'x' to the power 'n'
Select correct option:

►12
►10

►14
►None of these

Question # 8 of 10 (Start time: 11:45:24 AM) Total Marks: 1
If f(x) = 8x^2 -2x +5 , then f ‘ (0) will be…….
Select correct option:

►4
►-2
►-3
►-1

Question # 9 of 10 (Start time: 11:46:54 AM) Total Marks: 1
If the function is differentiable at a point then it is also ……..at that point
Select correct option:

►***Continuous***
►Not defined

Question # 10 of 10 (Start time: 11:48:06 AM) Total Marks: 1
If f(x)=3X2+2,then f'(x) will be_____. (Note: X2 means 'x' raised to the power 2)
Select correct option:

►3x+2
►6x
►6x+2
►None of these

Question # 1 of 10 (Start time: 11:35:52 AM) Total Marks: 1
If y=f(x) then instantaneous rate of change of y with respect to x at the point x0 is given by slope of tangent line at (x0,f(x0))
Select correct option:

►True
►False

Question # 2 of 10 (Start time: 11:37:09 AM) Total Marks: 1
If a secant line is drawn between two points P and Q on a curve, then the slope of this secant line is
Select correct option:

►=f(x2)-f(x1)
►=f(x2)-f(x1)/x2
►={f(x2)-f(x1)}/(x2-x1)
►=f(x1)-f(x2)/x2-x1

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►-23
►25
►-25

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If (0,0) and (2,2) are any two pints on a curve then the slope of a line parallel to the secant segment through these points is
►1
►-1
►2
►-2

Question # 5 of 10 (Start time: 11:41:26 AM) Total Marks: 1
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Select correct option:

►no where
►every where
►some where
►***none of these***

Question # 6 of 10 (Start time: 11:42:54 AM) Total Marks: 1
The process of finding the derivative is called_____
Select correct option:

►***Differentiation***
►Integration
►Function
►None of these

Question # 7 of 10 (Start time: 11:43:56 AM) Total Marks: 1
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Select correct option:

►12
►10
►14
►None of these

Question # 8 of 10 (Start time: 11:45:24 AM) Total Marks: 1

If f(x) = 8x^2 -2x +5 , then f ‘ (0) will be…….
Select correct option:

►4
►-2
►-3
►-1

Question # 9 of 10 (Start time: 11:46:54 AM) Total Marks: 1
If the function is differentiable at a point then it is also ……..at that point
Select correct option:

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►Not defined

10 of 10 (Start time: 11:48:06 AM) Total Marks: 1
If f(x)=3X2+2,then f'(x) will be_____. (Note: X2 means 'x' raised to the power 2)
Select correct option:

►3x+2
►6x
►6x+2
►None of these

Area of a rectangle whose width is 5 units and length is 6 units will be ….

Select correct option:

11 units

22 units

Question # 4 of 10 (Start time: 09:02:47 PM) Total Marks: 1

If Newton’s Method succeeded to get the approximate solution of an equation, then which of the following is NOT true about it.

Select correct option:

The slope of the tangent line (at any approximated point) must be non zero.

The tangent line (at any approximated point) is not parallel to x-axis.

The sequence of approximated points not convergent to the exact solution

None of these.

Question # 5 of 10 (Start time: 09:04:16 PM) Total Marks: 1

Integral of (1-2x) from [0,1] is ……..

Select correct option:

1

2

0

3

Question # 6 of 10 (Start time: 09:05:21 PM) Total Marks: 1

If f(x) = Cos(x) + x, then which of the following is NOT true about it.

Select correct option:

Its anti – derivative is Sin(x) + x^2/2 + 4.

Its anti – derivative is Sin(x) + x^2/2 + 6.

Its anti – derivative is Sin(x) + x^2/2 + 10.

Its anti – derivative is -Sin(x) + x^2/2 + 4.

Question # 7 of 10 (Start time: 09:06:54 PM) Total Marks: 1

Integral of 5^2 is NOTE: x^n means ‘x’ to the power ‘n’

Select correct option:

(1/3)5^3

10

25x

None of these

Question # 8 of 10 (Start time: 09:08:19 PM) Total Marks: 1

In the indefinite integral of x(y^2) w.r.t ‘y’ , the term ….. serve to identify the independent variable in the function.

Select correct option:

X

Y

dy

y^2

Question # 10 of 10 (Start time: 09:11:20 PM) Total Marks: 1

If f(x) = Sin (5x), then which of the following is NOT true about it.

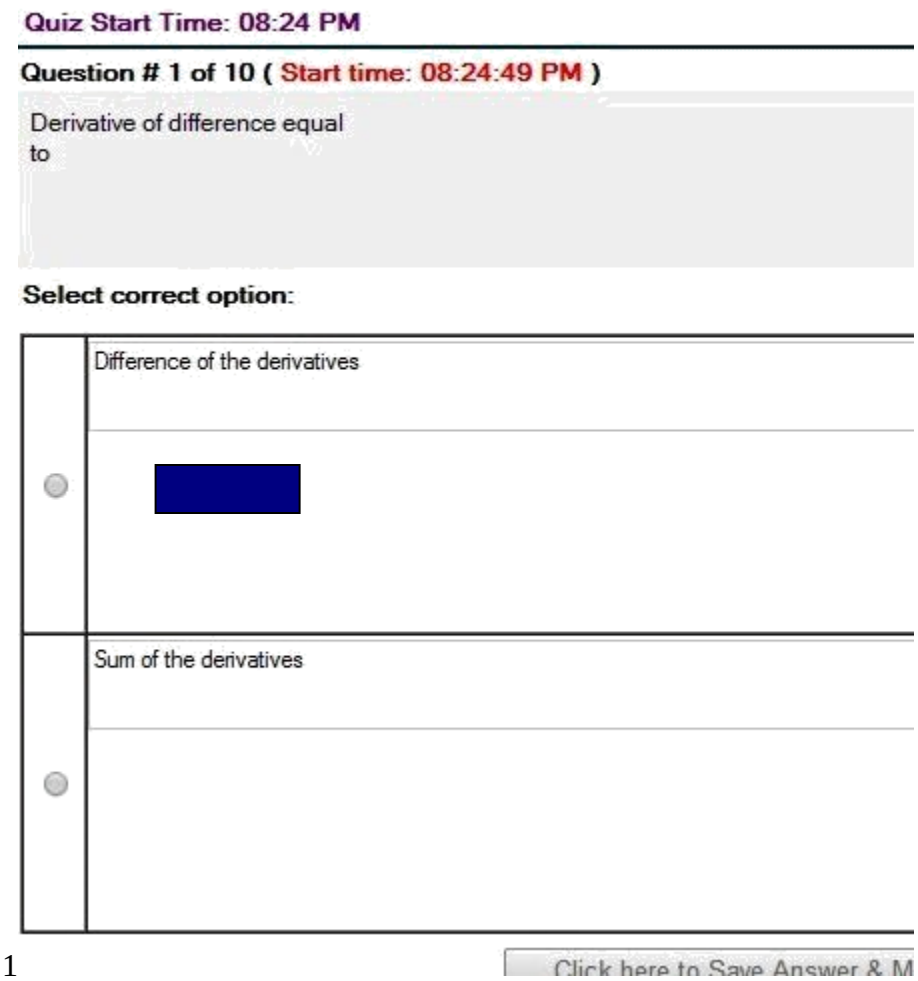
Select correct option:

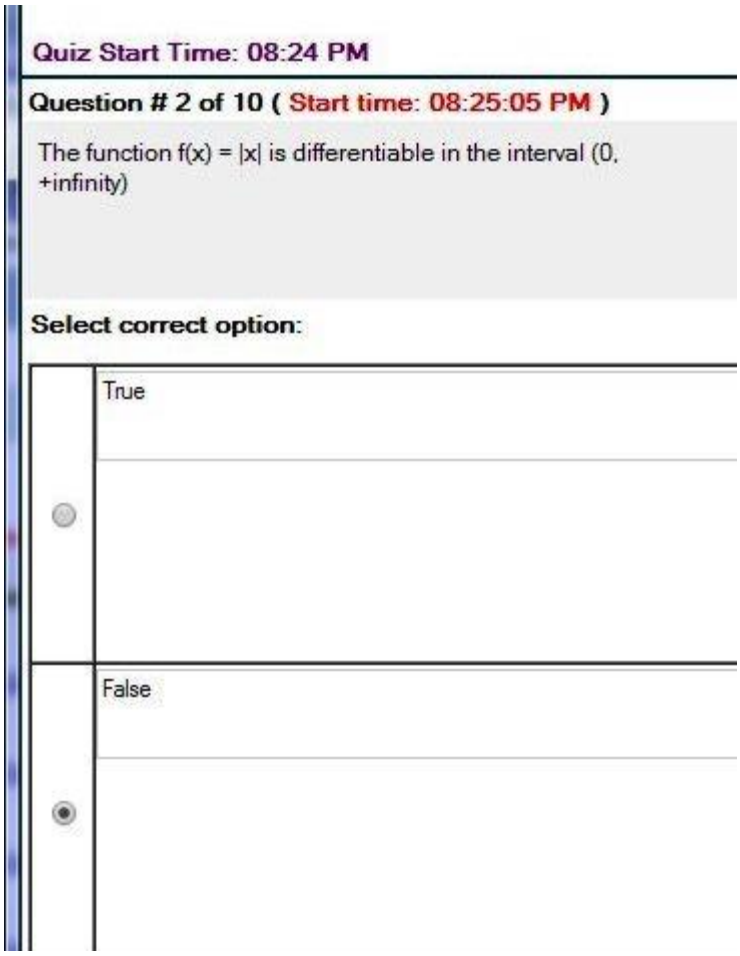
Its anti – derivative can be
–Cos (5x) + 3

Its anti – derivative can be
–Cos (5x) / 5 + 4.

Its anti – derivative can be
–Cos (5x) / 5 + 6.

Its anti – derivative can be
–Cos (5x) / 5 + 10.




Quiz Start Time: 08:24 PM
Question # 2 of 10 (Start time: 08:25:05 PM)
The function f(x) = |x| is differentiable in the interval (0, +infinity)
Select correct option:
True
False

Quiz Start Time: 08:24 PM

Question # 3 of 10 (Start time: 08:25:30 PM) **Total Ma**

If y=x^2 then average rate of change of y with respect to x over the interval [1, 2] is NOTE: x^n means 'x to the power 'n'

Select correct option:

○	3
○	2
○	1
○	None of these

Quiz Start Time: 08:24 PM

Question # 4 of 10 (Start time: 08:26:26 PM)

What is the derivative of

$$39x^{\frac{14}{13}}$$

Select correct option:

○	$42x^{\frac{1}{13}}$
○	$24x^{\frac{-1}{13}}$
○	$42x$
○	None of these

Quiz Start Time: 08:24 PM

Question # 5 of 10 (Start time: 08:27:44 PM)

Derivative exists only at the points where limit_____.

Select correct option:

- Does not exist
- Exist
- May or may not exist
- None of these

Quiz Start Time: 08:24 PM

Question # 6 of 10 (Start time: 08:27:57 PM)

On the geometrical figure like square, the points where the derivative does not exist are ------

Select correct option:

○	midpoints of its diagonals
○	midpoints of its sides
○	corners
○	the points of trisection of its sides

Quiz Start Time: 08:24 PM

Question # 7 of 10 (Start time: 08:28:36 PM)

On the straight line, the tangent at any point coincide with line --
---

Select correct option:

- no where
- every where
- some where
- none of these

question no 8

Quiz Start Time: 08:24 PM

Question # 8 of 10 (Start time: 08:29:01 PM)

If y=f(x) then the average rate of change of y with respect to x over the interval [x0,x1] is slope of secant line joining points (x0,f(x0)) and (x1
graph of f.

Select correct option:

- True
- False

Quiz Start Time: 08:24 PM

Question # 10 of 10 (Start time: 08:29:41 PM)

If f(x) = sec(x) and g(x) = tan(2x), then the derivative of f(x) - g(x) is _____. NOTE:- where x^n denotes the power of x.

Select correct option:

- sec(x) - 2tan(x).
- sec(x) - 2tan(2x).
- sec(x) + 2tan(x).
- sec(x)tan(x) - 2sec^2(2x).

Question # 1 of 10 (Start time: 11:06:59 PM)

Let y = x^2+1 Then the instantaneous rate of change of y with respect to x at the point -2 is

Select correct option:

○	-4
○	0
○	3
○	-1

Question # 2 of 10 (Start time: 11:08:26 PM)

Derivative of

$3x^7 + 4x^2 + 9$

is .....

Select correct option:

○	$21x^6 + 8x$
○	$21x^6 + 8x + 9$
○	$21x^6 + 4x^2 + 9$
○	None of these

Question # 7 of 10 (Start time: 11:13:44 PM)

If

$$f(x) = x^5 + 10$$

, then which of the following is true about it?

Select correct option:

○	Its derivative is x^4
○	Its derivative is $5x^4$
○	Its derivative is 10.
○	None of these.

Question # 10 of 10 (Start time: 11:16:49 PM)

What is the derivative of

$$39x^{\frac{14}{13}}$$

Select correct option:

○	$42x^{\frac{1}{13}}$
○	$24x^{\frac{-1}{13}}$
○	$42x$
○	None of these